

# BACKLOG BUDDY ? STUDY MATERIAL

Engineering Mechanics ? Quick Revision & Formula Sheet

Subject Code: ME105 Semester: 2

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Last-day revision guide with key definitions, formulas and diagrams checklist.

## Unit 1: Statics of Particles

- Definition, key concepts and significance of Statics of Particles
- Important diagrams / derivations / algorithms for Statics of Particles
- Solved examples and previous-year questions on Statics of Particles

## Unit 2: Equilibrium of Rigid Bodies

- Definition, key concepts and significance of Equilibrium of Rigid Bodies
- Important diagrams / derivations / algorithms for Equilibrium of Rigid Bodies
- Solved examples and previous-year questions on Equilibrium of Rigid Bodies

## Unit 3: Centroid and Moment of Inertia

- Definition, key concepts and significance of Centroid and Moment of Inertia
- Important diagrams / derivations / algorithms for Centroid and Moment of Inertia
- Solved examples and previous-year questions on Centroid and Moment of Inertia

## Unit 4: Dynamics of Particles

- Definition, key concepts and significance of Dynamics of Particles
- Important diagrams / derivations / algorithms for Dynamics of Particles
- Solved examples and previous-year questions on Dynamics of Particles

## Unit 5: Friction and Virtual Work

- Definition, key concepts and significance of Friction and Virtual Work
- Important diagrams / derivations / algorithms for Friction and Virtual Work
- Solved examples and previous-year questions on Friction and Virtual Work

Exam Tips:

- \* Start with the latest syllabus and exam pattern ? focus on units with the highest weightage first.
- \* Solve at least 3 previous year papers under timed conditions before the exam.
- \* Focus on frequently repeated topics ? they are strong preparation priorities based on available papers (not guarantees).
- \* Revise definitions, diagrams and standard derivations ? they earn easy marks.
- \* Practise numerical problems from every unit; papers usually include 2-3 compulsory problems.

Disclaimer: Sample study material for preparation support only.

**Appearance counts are based on available papers and are not exam predictions.**